

CROSBY, United Kingdom

WMO: 033160

Lat: 53.4972N

Lon: 3.0578W

Elev: 30

StdP: 14.68

Time Zone: 0.00 (EUW)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	26.1	29.2	21.5	16.1	29.9	24.1	18.3	32.0	46.4	45.9	42.2	46.4	5.4	50	0.708

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.9	76.2	65.1	72.3	63.4	69.3	62.1	66.6	73.2	64.8	70.1	63.3	67.9	10.6	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
64.3	90.3	68.8	62.6	85.1	67.3	61.2	80.9	66.0	31.2	73.4	29.8	70.2	28.8	68.1	80.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
37.8	33.1	29.4	DB	21.0	83.1	3.9	4.1	18.2	86.1	15.9	88.5	13.7	90.8	10.9	93.8	(4)
			WB	20.1	71.4	3.9	3.9	17.3	74.2	15.1	76.4	12.9	78.6	10.0	81.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	51.0	41.5	41.7	44.2	48.4	53.6	58.5	62.1	61.9	57.9	52.8	46.7	42.2
	DBStd	8.91	5.29	5.08	4.75	4.89	4.80	4.11	3.80	3.43	3.67	4.78	5.27	6.32
	HDD50	1179	265	233	185	85	18	0	0	0	1	27	120	245
	HDD65	5142	729	651	645	499	354	200	109	109	213	377	549	707
	CDD50	1556	1	2	6	37	131	256	374	369	239	115	22	4
	CDD65	39	0	0	0	0	1	5	18	13	2	0	0	0
	CDH74	221	0	0	0	0	14	39	105	57	6	0	0	0
	CDH80	34	0	0	0	0	0	4	20	10	0	0	0	0
Wind	WSAvg	13.4	14.8	14.6	13.3	12.2	12.5	12.4	12.6	12.9	12.9	13.8	14.3	14.5
Precipitation	PrecAvg	31.7	2.7	2.1	1.9	2.1	2.2	2.4	2.5	2.8	2.9	3.5	3.4	3.3
	PrecMax	43.2	5.0	3.9	4.4	4.3	4.3	7.1	5.0	5.0	5.6	6.7	7.3	6.1
	PrecMin	23.4	0.2	0.5	0.4	0.6	0.4	0.8	0.7	0.7	0.6	1.1	2.0	1.2
	PrecStd	4.6	1.3	0.8	0.9	1.1	0.9	1.5	1.1	1.0	1.3	1.4	1.2	1.3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	53.5	55.3	60.1	69.6	75.8	79.5	82.8	80.5	73.8	65.9	59.0	55.5
		MCWB	50.6	50.0	52.1	56.9	63.0	64.7	67.1	67.0	63.6	59.9	56.1	52.3
	2%	DB	51.2	52.0	55.4	62.9	69.7	72.7	76.3	73.7	69.2	62.8	57.1	53.6
		MCWB	48.8	48.9	50.4	54.0	60.1	62.8	65.5	65.0	61.7	58.3	54.5	50.8
	5%	DB	49.8	50.2	52.7	58.8	64.8	68.6	72.2	69.9	66.3	61.1	55.2	51.9
		MCWB	47.5	47.4	48.9	52.2	57.1	61.1	63.9	63.3	60.5	57.3	52.6	49.3
	10%	DB	48.4	48.7	50.7	55.9	61.3	65.6	68.9	67.5	64.2	59.5	53.6	50.0
		MCWB	46.4	46.1	47.7	50.8	55.3	59.6	62.3	62.1	59.0	56.0	51.2	47.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	51.2	51.4	53.7	58.3	65.2	67.8	69.0	69.0	65.8	61.2	56.6	53.0
		MCDB	52.8	53.8	57.9	65.5	73.0	72.6	78.3	76.3	71.2	64.3	58.4	54.9
	2%	WB	49.4	49.3	51.2	55.2	61.1	64.2	66.5	66.2	63.4	59.5	54.9	51.2
		MCDB	50.9	51.5	54.3	60.8	68.0	70.9	74.1	71.7	67.3	62.1	56.7	53.2
	5%	WB	48.0	47.8	49.6	53.0	58.4	61.9	64.7	64.6	61.5	58.0	53.0	49.5
		MCDB	49.4	49.8	51.9	57.8	63.3	67.2	70.7	68.6	64.9	60.5	54.7	51.4
	10%	WB	46.6	46.5	48.2	51.3	56.2	60.2	63.0	63.1	60.0	56.5	51.6	48.0
		MCDB	48.1	48.5	50.5	55.2	60.3	64.9	67.9	66.6	63.2	58.9	53.3	49.8
Mean Daily Temperature Range	5% DB	MDBR	7.1	8.2	9.7	12.0	12.4	11.5	10.9	10.1	10.7	9.5	7.8	7.9
		MCDBR	7.3	8.9	12.7	17.8	19.8	18.8	18.7	15.3	14.8	10.4	7.4	7.8
		MCWBR	6.9	7.2	9.1	11.3	12.1	11.8	10.4	9.2	9.4	7.6	6.8	7.2
	5% WB	MCDBR	7.1	8.1	10.8	15.2	17.7	16.0	16.2	12.9	12.4	9.0	7.0	7.6
		MCWBR	7.1	7.1	8.4	10.6	12.0	11.1	9.8	8.7	9.2	7.5	6.8	7.4
Clear-Sky Solar Irradiance	taub	0.330	0.343	0.372	0.394	0.398	0.395	0.395	0.381	0.369	0.352	0.328	0.318	
	taud	2.314	2.373	2.306	2.265	2.301	2.340	2.345	2.399	2.432	2.436	2.379	2.288	
	Ebn at Noon	199	239	255	264	270	272	269	266	255	233	200	177	
	Edh at Noon	19	25	32	38	38	37	36	33	28	23	18	16	
All-Sky Solar Radiation	RadAvg	199	416	784	1214	1548	1618	1533	1260	895	496	243	150	
	RadStd	20	45	61	107	113	146	148	80	66	42	25	15	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (65 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page